

Pancancer Survival Prediction using Deep Learning with Multimodal Integration

A deep learning architecture combining unsupervised representation learning and attention-based multimodal fusion for pancancer survival prognosis.

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Abstract

Survival prognosis is integral to personalized oncology. We present a multimodal deep learning model that leverages unsupervised representation learning and an attention-based fusion layer to extract and weight salient signals across clinical, mRNA, miRNA, and CNV data. Trained on TCGA pancancer cohorts (6,656 train), the model reaches training C-index up to 0.928 and shows improved prediction across most cancer types. Our approach demonstrates that deep representational learning with attention can synthesize complex multi-omic inputs toward clinically useful risk scores.

Motivation



The Reality

High cancer mortality—accurate prognosis improves treatment timing and outcomes.



Opportunity

Multi-omics (genomics, transcriptomics, CNV, miRNA) provide complementary biological insight.



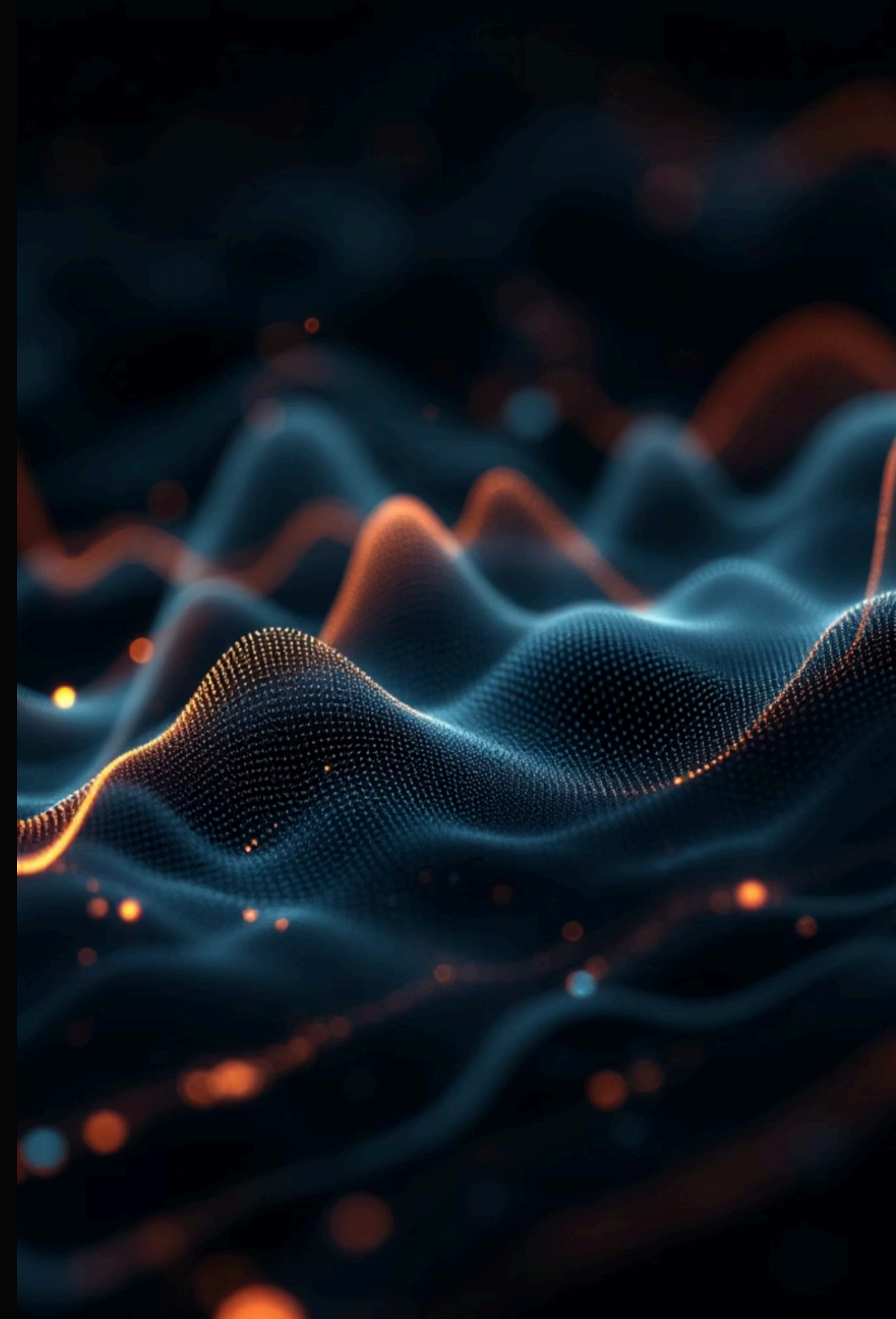
Gap

Most models use single modalities and struggle to model cross-omic interactions and noise.



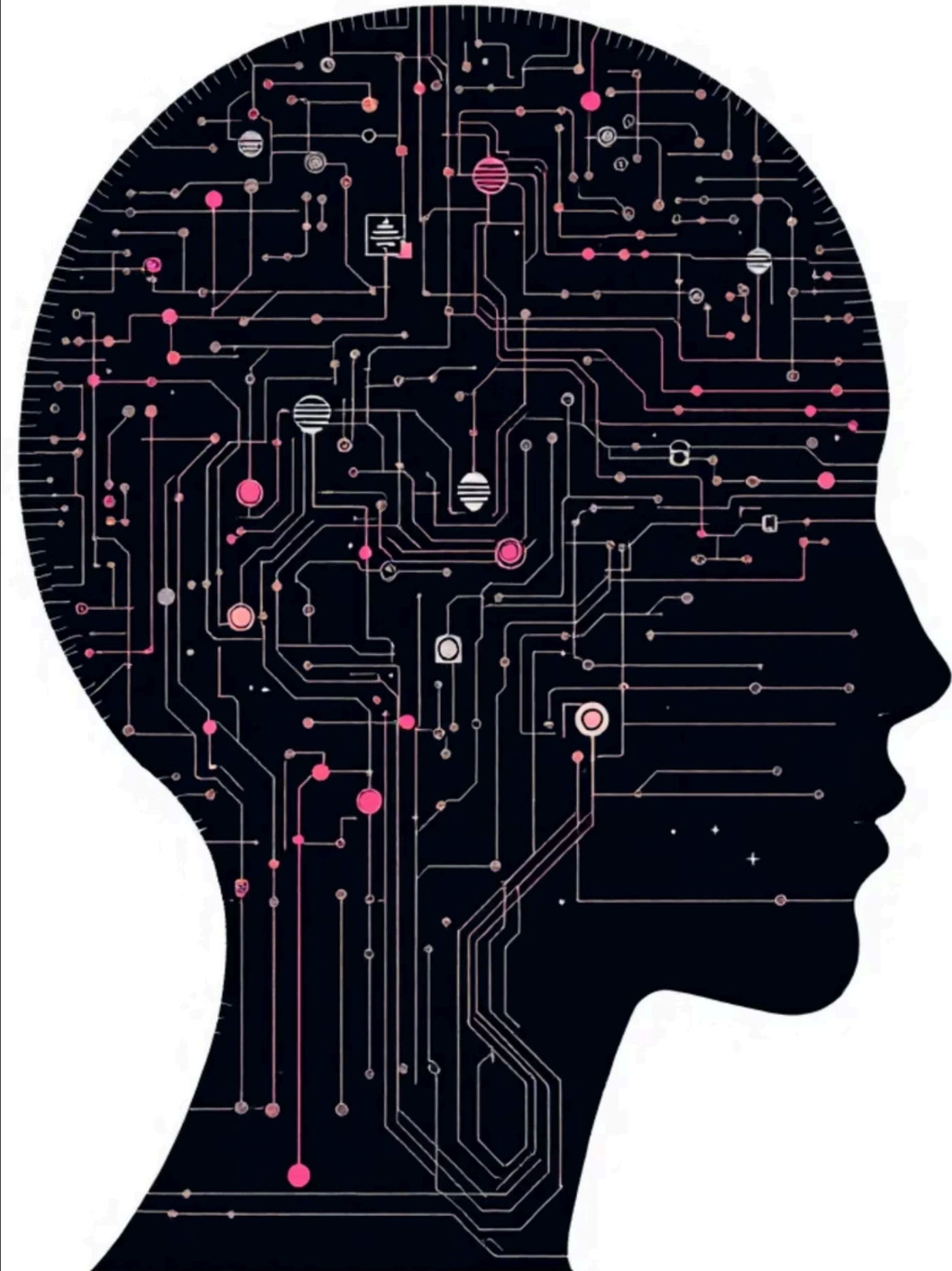
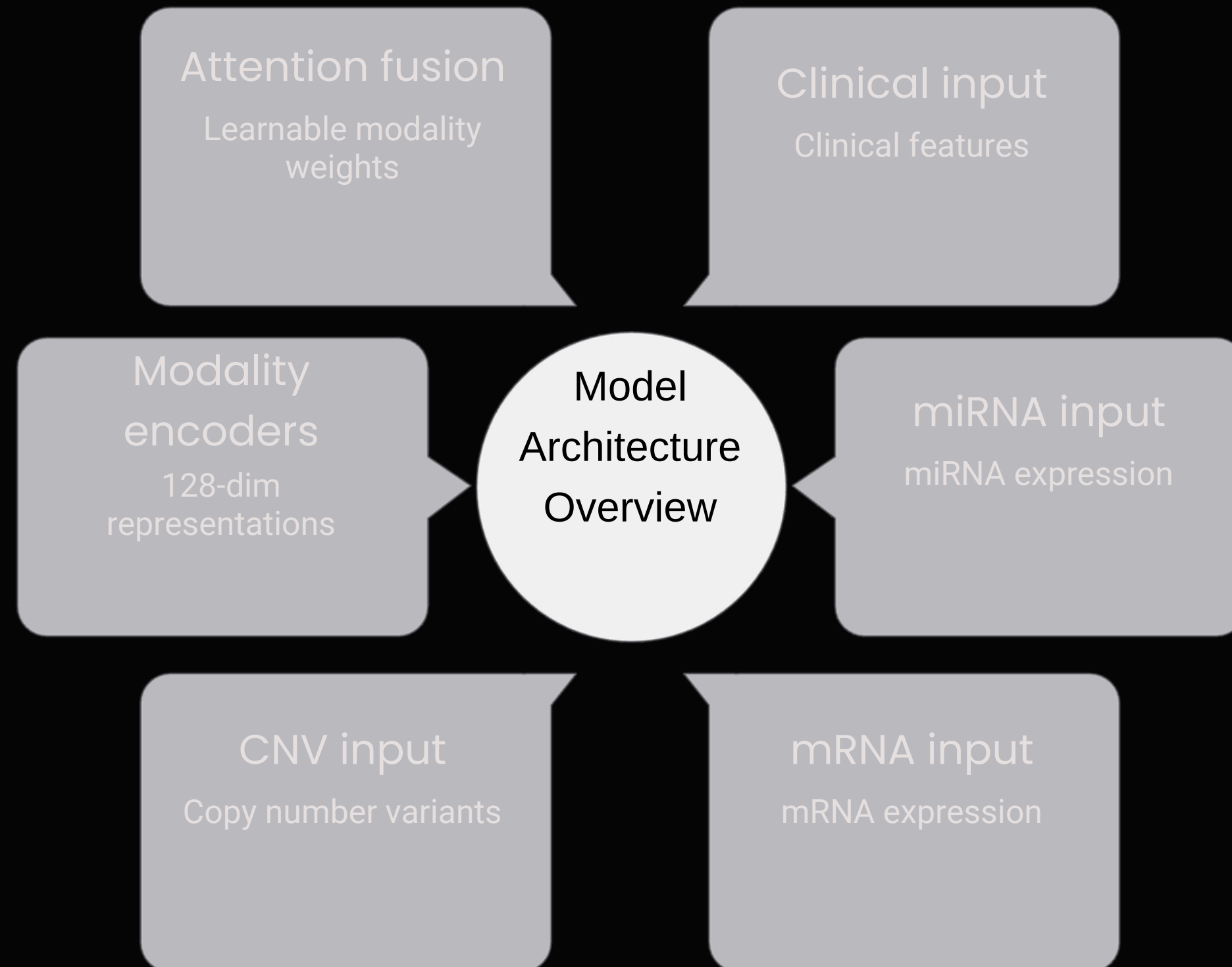
Why DL & Attention?

Nonlinear feature extraction + attention enables modality weighting and robust integration of noisy, high-dimensional data.



Methodology – Model Architecture (Overview)

The model has three components: modality-specific unsupervised encoders producing 128-dim representations, attention-based fusion (learnable weights normalized by softmax), and a two-layer survival predictor (Cox loss + $\lambda \times$ Siamese similarity loss). Training optimizes representation quality and survival discrimination jointly.





Dataset & Preprocessing

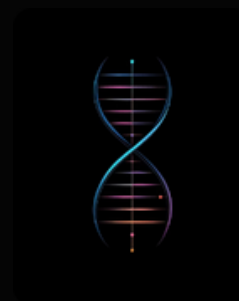
Source: TCGA PanCanAtlas. Dataset split 60/20/20 → Train: 6,656, Val: 2,219, Test: 2,219. Modalities and feature reduction:
Preprocessing: variance thresholding, min-max normalization [0,1], missing modalities → zero vectors.



Clinical
5 variables (4 categorical + age) → embedded to 128-dim



mRNA
20,531 genes → variance threshold → 1,579 features



miRNA
743 features retained



CNV
24,776 genes → 2,711 features; 8.3% missing → zero-vector imputation

Methodology — Representation Learning & Loss

Modality-specific encoder architectures (all with Dropout=0.4 and BN):

Clinical

Embed categories

→ BN → FC → 128

miRNA

743 → 400 → 128

mRNA

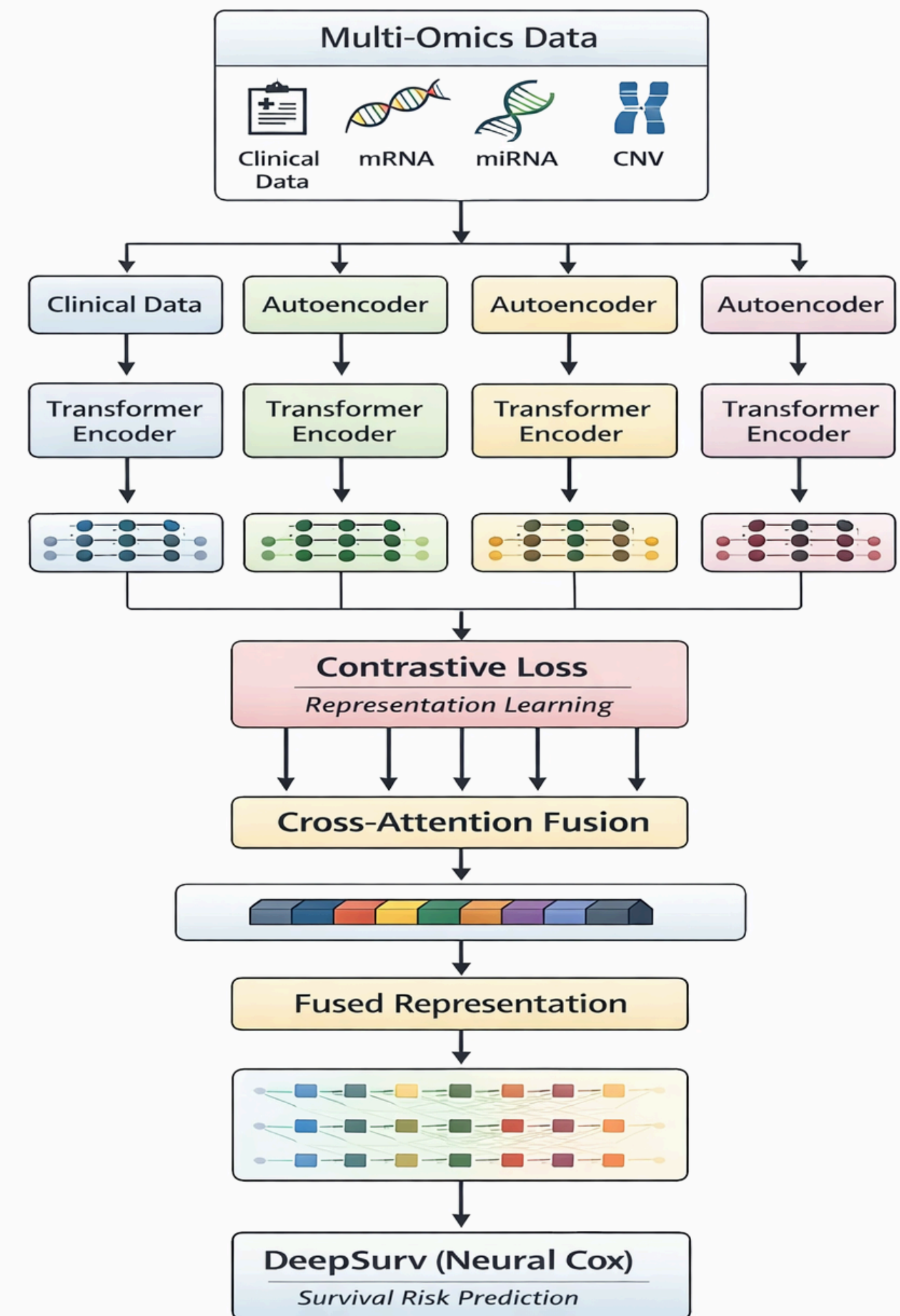
1,579 → 800 →

400 → 128

CNV

2,711 → 1,300 → 700 → 300 → 128

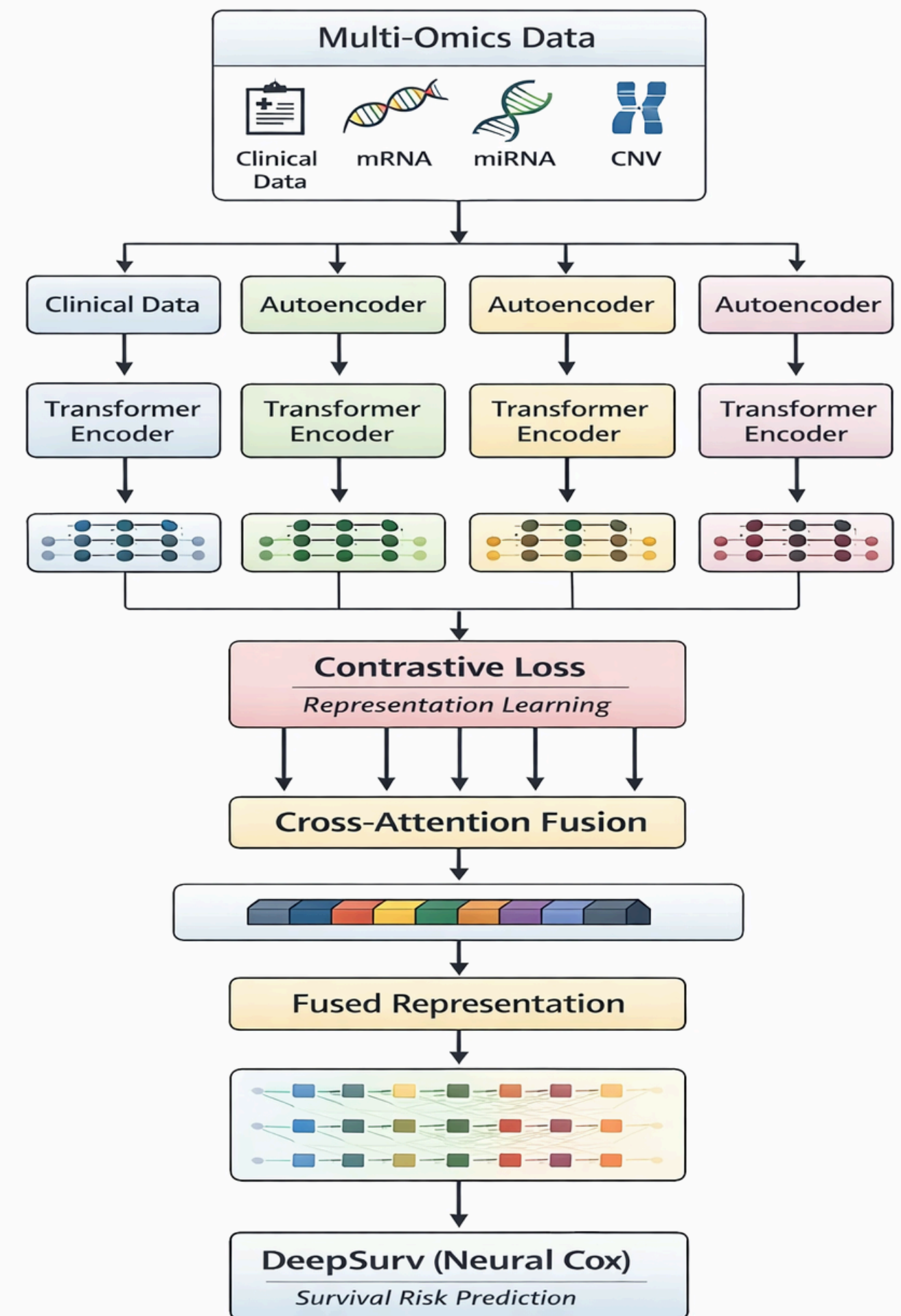
Modified Siamese similarity loss: random pair matching within batch reduces pairwise complexity from $O(n^2)$ to $O(n)$. Similarity uses cosine-like inner product normalized by vector norms; margin $M=0.2$ to prevent collapse.



What we Approach:

1. Siamese → Contrastive Loss
2. Attention → Multi-head Attention
3. Cox → DeepSurv

Autoencoder → Transformer → Contrastive Learning → Multi-Head
Attention → DeepSurv



Training & Experimental Results

Implementation: PyTorch/TensorFlow. Hyperparameters: batch=250, lr=0.01, epochs=30, dropout=0.4. Representative runs:

Run 1 (summary)

Train C-index improved to **0.926** by epoch 30; best val C-index 0.7573.

Run 2 (summary)

Train C-index reached **0.928**; validation peaked early (epoch 4–8), best val C-index $\approx$ **0.7835** $\rightarrow$ overfitting observed.

Model vs Baselines (C-index)

Our model outperforms CPH, RSF, and MultiSurv across most modalities. Clinical+mRNA: **0.780** (best). All-modality fusion: 0.772.

Per-cancer highlights

THCA: **0.943** (best); several cancers show strong gains vs prior work.



Conclusions, Limitations & Next Steps

Conclusions

Unsupervised representations + attention fusion enable robust multimodal survival prediction. Clinical+mRNA combination was optimal; modified Siamese loss improved training efficiency.

Limitations

Missing-data imputation via zero vectors may bias results; overfitting observed; no external validation yet; CNV sometimes adds noise.

Immediate Next Steps

- Implement early stopping and L2 regularization
- Evaluate lower/higher representation dims (64, 256, 512)
- External validation on independent cohorts
- Explore transfer learning for modality imputation and cross-attention

Impact

Improved pancancer prognostication can guide treatment stratification, enable earlier intervention, and advance personalized oncology.



References & Resources

Fan et al. (2023), Cheerla & Gevaert (2019), TCGA PanCanAtlas, UCSC Xena.

Code (base paper): <https://github.com/ZhangqiJiang07/MultimodalSurvivalPrediction>